

**MANIPAL UNIVERSITY JAIPUR**  
Department Artificial Intelligence and Machine Learning  
Course Hand-out

**A. Basic Details:**

|                                                     |                                                                                                                                                                                                                              |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Programme Name:                                     | BTech CSE(AI&ML)                                                                                                                                                                                                             |
| Course Name:                                        | Machine Learning                                                                                                                                                                                                             |
| Course Code:                                        | AIM3101                                                                                                                                                                                                                      |
| LTPC ( <i>Lecture Tutorial Practical Credits</i> ): | 3 1 0 4                                                                                                                                                                                                                      |
| Session:                                            | Jul-Nov 2026                                                                                                                                                                                                                 |
| Class:                                              | V                                                                                                                                                                                                                            |
| Course Coordinator:                                 | Dr. Anamika Dhillon                                                                                                                                                                                                          |
| Course Instructor(s):                               | Dr. Anamika Dhillon / Dr. Surendra Solanki / Mr. Harish Sharma / Dr. Pranshu Pranjali / Dr. Anamika Dhillon / Dr. Pratima Kumari / Dr. Sonia Choudhary / Dr. Jayant Mahawar / Dr. Pratima Kumari / Dr. Yadendra Pratap Singh |
| Adjunct Faculty                                     | Dr. Suman Ghosh                                                                                                                                                                                                              |

**B. Introduction** This course on Machine Learning provides a comprehensive overview of foundational concepts and techniques in the field. It covers the basics of labeled and unlabeled data, types of learning paradigms (supervised, unsupervised, parametric, and non-parametric models), and essential learning theory, including the bias-variance trade-off and model selection. Students will explore supervised learning methods for regression and classification, delve into ensemble techniques like bagging and boosting, and gain insights into unsupervised learning approaches for clustering and dimensionality reduction. The course also emphasizes performance evaluation metrics to build effective and robust machine learning models.

**C. Course Outcomes:**

| <i>CO Statement</i>                                                                                                                                                                                                                       | <i>CO</i> | <i>Level</i>          | <i>Target Attainment %</i> | <i>Target Attainment level</i> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------|----------------------------|--------------------------------|
| Explain the fundamental concepts of machine learning, including labelled and unlabelled data, types of machine learning, supervised vs. unsupervised learning, parametric vs. non-parametric models, and learning theory.                 | AIM3101.1 | Cognitive: Understand | ≥ 80                       | Level 3                        |
| Apply regression techniques such as linear regression, gradient descent, ridge regression, lasso regression, and logistic regression to solve supervised learning problems.                                                               | AIM3101.2 | Cognitive: Apply      | ≥ 70% < 80%                | Level 2                        |
| Analyze and evaluate classification methods, including LDA, k-NN, SVMs, Bayesian networks, naïve Bayes, decision trees, and Gaussian mixture models, using performance metrics like confusion matrix, F1 score, accuracy, and ROC curves. | AIM3101.3 | Cognitive: Evaluate   | ≥ 80                       | Level 3                        |
| Design and implement ensemble learning models such as bagging, boosting (e.g., AdaBoost, XGBoost), and random forests, and validate models using techniques like K-fold cross-validation.                                                 | AIM3101.4 | Cognitive: Create     | ≥ 70% < 80%                | Level 2                        |
| Examine clustering methods (e.g., k-means, hierarchical clustering) and dimensionality reduction techniques (e.g.,                                                                                                                        | AIM3101.5 | Cognitive: Analyze    | ≥ 70% < 80%                | Level 2                        |

|                                                                       |  |  |  |  |
|-----------------------------------------------------------------------|--|--|--|--|
| PCA, factor analysis) for analyzing and interpreting unlabelled data. |  |  |  |  |
|-----------------------------------------------------------------------|--|--|--|--|

#### D. Program Outcomes and Program Specific Outcomes:

**[PO.1] Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering discipline to the solution of complex engineering problems.

**[PO.2] Problem Analysis:** Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

**[PO.3] Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

**[PO.4] Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

**[PO.5] Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

**[PO.6] The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

**[PO.7] Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

**[PO.8] Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practices.

**[PO.9] Individual and Teamwork:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

**[PO.10] Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

**[PO.11] Project Management and Finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

**[PO.12] Life-long Learning:** Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

**PROGRAM SPECIFIC OUTCOMES:**

**[PSO.1]** Graduates will be able to examine the applications of Artificial Intelligence and Machine Learning in real-life problems.

**[PSO.2]** Graduates will be able to design and implement intelligent systems for multidisciplinary problems.

**E. Assessment Plan**

| Criteria                           | Description                                                                                                                                                                             | Maximum Marks |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Internal Assessment<br>(Summative) | Mid-Term Examination<br>(Close Book)                                                                                                                                                    | 30            |
|                                    | Quiz (2): 10 Marks<br>Assignments (2): 10 Marks<br>Presentation (max 2 student):<br>10 Marks                                                                                            | 30            |
| End Term Exam<br>(Summative)       | End Term Exam<br>(Close Book)                                                                                                                                                           | 40            |
|                                    | Total                                                                                                                                                                                   | 100           |
| Attendance<br>(Formative)          | A minimum of 75% Attendance is required to be maintained by a student to be eligible to appear for the ETE. The allowance of 25% includes all types of leaves including medical leaves. |               |

**F. Syllabus:**

**Introduction to Machine Learning:** Basics of Machine Learning, labelled, unlabelled and, Types of Machine Learning: supervised, unsupervised, semi-supervised and reinforcement learning, parametric vs non-parametric models, learning theory, bias-variance tradeoff, model selection, Data Visualization Concepts, Data Sampling Techniques.

**Regression Techniques and Evaluation:** Linear Regression, Gradient descent for Linear Regression, Ridge and Lasso Regression. Logistic Regression. Performance measures for Regression models. Cost functions, Underfitting, Overfitting.

**Core Classification Algorithms:** k Nearest Neighbours (KNN), Support Vector Machines: kernel functions, Naïve Bayes, Decision Trees. Performance & Evaluation of Classifiers: Confusion matrix, F1 score, Accuracy, Recall, Precision, Precision-recall curve, ROC curve.

**Advanced Classification and Ensemble Techniques:** Ensemble Learning: Concept of weak learners, Bagging and Boosting, Adaptive Boosting, Extreme Gradient Boosting (XGBoost), Random Forests, Bayesian Networks, Model Validation: K-Fold and Cross-Validation, holdout method, Gaussian Mixture Models (GMM).

**Unsupervised learning:** Clustering: Partitioning, Hierarchical and Density based methods, Agglomerative Clustering, K-Means, K-Medoids. Association Rule Mining. Dimensionality Reduction: Feature Extraction and Feature Selection. Principal Component Analysis (PCA), Partial Least Squares, Latent Factors, Factor Analysis.

### **Textbooks**

Machine Learning by Dutt. Published by Pearson. Relevant to Computer Science.

Tom M. Mitchell - "Machine Learning," McGraw-Hill Education, 1997. A foundational textbook introducing the principles and algorithms of machine learning.

Christopher M. Bishop - "Pattern Recognition and Machine Learning," Springer, 2006. Covers both the theoretical and practical aspects of pattern recognition and machine learning.

### **Reference Books**

Ian Goodfellow, Yoshua Bengio, and Aaron Courville - "Deep Learning," MIT Press, 2016. Provides an in-depth understanding of deep learning techniques and their applications.

Ethem Alpaydin - "Introduction to Machine Learning," MIT Press, 4th Edition, 2020. Offers a concise introduction to various machine learning methods and their real-world implementations.

### **G. Lecture Plan:**

| Lecture No. | Topic                        | Session Outcome                                     | Corresponding CO | Mode of delivery                | Mode of assessing CO |
|-------------|------------------------------|-----------------------------------------------------|------------------|---------------------------------|----------------------|
| 1           | Basics of Machine Learning   | Understand the fundamentals of machine learning.    | AIM3101.1        | Lecture-based teaching-learning | Short Quiz           |
| 2           | Labelled and Unlabelled Data | Differentiate between labelled and unlabelled data. | AIM3101.1        | Technology-based learning       | Online Assignment    |
| 3           | Types of Machine Learning    | Learn about supervised, unsupervised, and           | AIM3101.1        | Lecture-based teaching-learning | Quiz                 |

|    |                                             |                                                                    |           |                                  |                            |
|----|---------------------------------------------|--------------------------------------------------------------------|-----------|----------------------------------|----------------------------|
|    |                                             | reinforcement learning.                                            |           |                                  |                            |
| 4  | Supervised vs. Unsupervised Learning        | Compare supervised and unsupervised learning techniques.           | AIM3101.1 | Lecture-based teaching-learning  | Conceptual Questions       |
| 5  | Parametric vs. Non-Parametric Models        | Identify differences between parametric and non-parametric models. | AIM3101.1 | Lecture-based teaching-learning  | Assignment                 |
| 6  | Learning Theory and Bias-Variance Trade-Off | Explore bias-variance trade-off and its role in performance.       | AIM3101.1 | Lecture-based teaching-learning  | Quiz                       |
| 7  | Model Selection                             | Understand methods for optimal model selection.                    | AIM3101.1 | Lecture-based teaching-learning  | Online Quiz                |
| 8  | Introduction to Regression Models           | Learn the basics of regression in supervised learning.             | AIM3101.2 | Lecture-based teaching-learning  | Short Quiz                 |
| 9  | Linear Regression                           | Understand linear regression and its applications.                 | AIM3101.2 | Lecture-based teaching-learning  | Problem-Solving Assignment |
| 10 | Gradient Descent for Linear Regression      | Explore gradient descent for optimizing regression models.         | AIM3101.2 | Learning through problem-solving | Practical Task             |
| 11 | Ridge and Lasso Regression                  | Differentiate between Ridge and Lasso regression techniques.       | AIM3101.2 | Lecture-based teaching-learning  | Assignment                 |
| 12 | Logistic Regression                         | Learn logistic regression for classification problems.             | AIM3101.2 | Lecture-based teaching-learning  | Quiz                       |
| 13 | Performance Measures for Regression Models  | Evaluate regression models using MSE, RMSE, etc.                   | AIM3101.2 | Technology-based learning        | Conceptual Questions       |
| 14 | Cost Functions, Underfitting, Overfitting   | Analyze cost functions, underfitting, and overfitting.             | AIM3101.2 | Lecture-based teaching-learning  | Quiz                       |

|    |                                            |                                                             |           |                                 |                            |
|----|--------------------------------------------|-------------------------------------------------------------|-----------|---------------------------------|----------------------------|
| 15 | Introduction to Classification             | Learn classification basics in supervised learning.         | AIM3101.3 | Lecture-based teaching-learning | Quiz                       |
| 16 | Linear Discriminant Analysis               | Understand LDA for classification.                          | AIM3101.3 | Lecture-based teaching-learning | Assignment                 |
| 17 | k-Nearest Neighbors (k-NN)                 | Explore k-NN and its applications.                          | AIM3101.3 | Lecture-based teaching-learning | Conceptual Questions       |
| 18 | Support Vector Machines (SVMs)             | Understand SVMs and kernel functions.                       | AIM3101.3 | Technology-based learning       | Problem-Solving Assignment |
| 19 | Bayesian Networks and Naïve Bayes          | Learn Bayesian methods for classification.                  | AIM3101.3 | Lecture-based teaching-learning | Quiz                       |
| 20 | Decision Trees and Gaussian Mixture Models | Explore decision trees and Gaussian mixture models.         | AIM3101.3 | Lecture-based teaching-learning | Short Quiz                 |
| 21 | Performance & Evaluation of Classifiers    | Evaluate classifiers using confusion matrix, F1 score, etc. | AIM3101.3 | Technology-based learning       | Assignment                 |
| 22 | Introduction to Ensemble Methods           | Learn the concept of weak learners and ensemble methods.    | AIM3101.4 | Flipped Classroom               | Quiz                       |
| 23 | Bagging and Random Forests                 | Understand bagging and random forests.                      | AIM3101.4 | Flipped Classroom               | Problem-Solving Assignment |
| 24 | Boosting and Adaptive Boosting             | Explore boosting techniques.                                | AIM3101.4 | Flipped Classroom               | Quiz                       |
| 25 | Extreme Gradient Boosting (XGBoost)        | Learn about XGBoost.                                        | AIM3101.4 | Flipped Classroom               | Conceptual Questions       |
| 26 | K-Fold Cross-Validation                    | Understand K-Fold cross-validation in ML.                   | AIM3101.4 | Flipped Classroom               | Quiz                       |
| 27 | Introduction to Clustering                 | Learn the basics of clustering in unsupervised learning.    | AIM3101.5 | Lecture-based teaching-learning | Assignment                 |

|    |                                          |                                                                  |           |                                 |                            |
|----|------------------------------------------|------------------------------------------------------------------|-----------|---------------------------------|----------------------------|
| 28 | K-Means Clustering                       | Understand K-Means clustering.                                   | AIM3101.5 | Flipped Classroom               | Practical Task             |
| 29 | K-Medoids Clustering                     | Explore K-Medoids clustering techniques.                         | AIM3101.5 | Flipped Classroom               | Quiz                       |
| 30 | Hierarchical Clustering                  | Analyze hierarchical clustering methods.                         | AIM3101.5 | Lecture-based teaching-learning | Conceptual Questions       |
| 31 | Density-Based Clustering (DBSCAN)        | Explore DBSCAN clustering.                                       | AIM3101.5 | Technology-based learning       | Problem-Solving Assignment |
| 32 | Association in Unsupervised Learning     | Learn association rules for pattern discovery.                   | AIM3101.5 | Expeditionary Learning          | Group Project              |
| 33 | Introduction to Dimensionality Reduction | Understand dimensionality reduction techniques.                  | AIM3101.5 | Lecture-based teaching-learning | Quiz                       |
| 34 | Principal Component Analysis (PCA)       | Analyze PCA for dimensionality reduction.                        | AIM3101.5 | Flipped Classroom               | Assignment                 |
| 35 | Partial Least Squares                    | Understand Partial Least Squares.                                | AIM3101.5 | Lecture-based teaching-learning | Conceptual Questions       |
| 36 | Factor Analysis and Comparison with PCA  | Explore factor analysis and compare with PCA.                    | AIM3101.5 | Lecture-based teaching-learning | Quiz                       |
| 37 | Feature Extraction vs. Feature Selection | Understand differences between feature extraction and selection. | AIM3101.5 | Technology-based learning       | Assignment                 |
| 38 | Deep Dive into PCA Techniques            | Explore advanced techniques in PCA.                              | AIM3101.5 | Flipped Classroom               | Problem-Solving Assignment |
| 39 | Singular Value Decomposition (SVD)       | Understand the SVD and its relation to PCA.                      | AIM3101.5 | Technology-based learning       | Quiz                       |
| 40 | Independent Component Analysis (ICA)     | Learn about ICA and its applications.                            | AIM3101.5 | Lecture-based teaching-learning | Assignment                 |

|                      |                                                  |                                                                    |                      |                                 |                           |
|----------------------|--------------------------------------------------|--------------------------------------------------------------------|----------------------|---------------------------------|---------------------------|
| 41                   | t-SNE for Dimensionality Reduction               | Explore t-SNE for data visualization and dimensionality reduction. | AIM3101.5            | Lecture-based teaching-learning | Practical Task            |
| 42                   | Comparison of Dimensionality Reduction Methods   | Compare various dimensionality reduction techniques.               | AIM3101.5            | Expeditionary Learning          | Group Project             |
| 43                   | Case Study 1: Predictive Analytics in Healthcare | Apply ML concepts to predict healthcare outcomes.                  | AIM3101.1, AIM3101.2 | Expeditionary Learning          | Presentations and Reports |
| 44                   | Case Study 2: Fraud Detection in Banking         | Analyze ML techniques used in fraud detection in banking systems.  | AIM3101.2, AIM3101.3 | Expeditionary Learning          | Presentations and Reports |
| 45                   | Case Study 3: Image Classification with CNNs     | Use Convolutional Neural Networks (CNNs) for image classification. | AIM3101.3, AIM3101.4 | Expeditionary Learning          | Presentations and Reports |
| 46                   | Case Study 4: Sentiment Analysis in Social Media | Apply ML to analyze sentiment in social media data.                | AIM3101.4, AIM3101.5 | Expeditionary Learning          | Presentations and Reports |
| 47                   | Case Study 5: Customer Segmentation in Marketing | Use clustering techniques for customer segmentation in marketing.  | AIM3101.5            | Expeditionary Learning          | Presentations and Reports |
| 48                   | Case Study 6: Recommendation Systems             | Build a recommendation system using collaborative filtering.       | AIM3101.5            | Expeditionary Learning          | Presentations and Reports |
| End Term Examination |                                                  |                                                                    |                      |                                 |                           |

## H. Course Articulation Matrix:

| CO         | STATEMENT                                                                                                                                                                                                            |             |             |             |             | CORRELATION WITH PROGRAM OUTCOMES |             |             |             |             |              |              |              |           |           |  |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------|-------------|-------------|-----------------------------------|-------------|-------------|-------------|-------------|--------------|--------------|--------------|-----------|-----------|--|
|            |                                                                                                                                                                                                                      | P<br>O<br>1 | P<br>O<br>2 | P<br>O<br>3 | P<br>O<br>4 | P<br>O<br>5                       | P<br>O<br>6 | P<br>O<br>7 | P<br>O<br>8 | P<br>O<br>9 | P<br>O<br>10 | P<br>O<br>11 | P<br>O<br>12 | PS<br>O 1 | PS<br>O 2 |  |
| AIM3101.1: | Explain fundamental ML concepts (labelled/unlabelled data, types, supervised/unsupervised learning, parametric/non-parametric models, and learning theory.)                                                          | 3           | 3           | 1           | 1           | 1                                 | 1           | 1           |             | 3           |              |              | 2            | 2         | 2         |  |
| AIM3101.2: | Apply regression techniques (linear regression, gradient descent, ridge, lasso, logistic regression) to solve supervised learning problems.                                                                          | 3           | 3           | 3           |             | 3                                 | 3           |             |             |             | 3            |              | 2            | 1         | 2         |  |
| AIM3101.3: | Analyze and evaluate classification methods (LDA, k-NN, SVMs, Bayesian networks, naïve Bayes, decision trees, Gaussian mixture models) using performance metrics (confusion matrix, F1 score, accuracy, ROC curves). | 3           | 3           | 3           | 3           | 3                                 |             | 3           |             | 3           | 3            | 3            | 2            | 2         | 2         |  |
| AIM3101.4: | Design and implement ensemble learning models (bagging, boosting, AdaBoost,                                                                                                                                          | 3           |             | 3           | 3           | 3                                 | 3           |             |             |             | 3            | 3            | 3            | 2         | 2         |  |

|            |                                                                                                                                                                   |  |   |   |   |  |   |   |  |   |   |   |   |   |   |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---|---|---|--|---|---|--|---|---|---|---|---|---|
|            | XGBoost, random forests) and validate using K-fold cross-validation.                                                                                              |  |   |   |   |  |   |   |  |   |   |   |   |   |   |
| AIM3101.5: | Examine clustering methods (k-means, hierarchical) and dimensionality reduction techniques (PCA, factor analysis) for analysing and interpreting unlabelled data. |  | 3 | 3 | 3 |  | 3 | 3 |  | 3 | 3 | 3 | 3 | 2 | 3 |

1: Low Correlation 2: Moderate Correlation 3: Substantial Correlation

**Course Co-Ordinator**

**Student Representative**

**Name:**

**Registration No.:**

**Head of Dept.**

**CSE(AIML)**